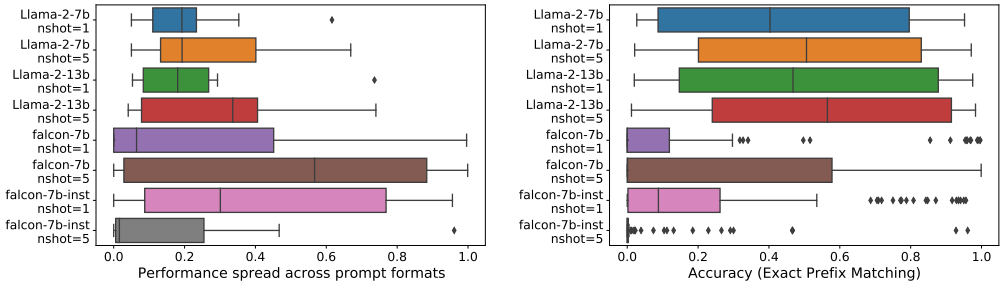

(a) Spreads across models and n-shots.

(b) Accuracy variance by model

Figure 18: Instruction Induction tasks' spreads and accuracy across models. Exact prefix matching is used as accuracy metric.

Experiments with continuous metrics in open-ended text generation tasks. Throughout the paper we focus on tasks with a single valid output, whether in classification tasks or in short-text generation tasks. This decision is intentional, since it guarantees that a variation in the metric truly represents a variation in model performance. We have shown that spread remains high when considering option ranking or exact prefix matching as accuracy metric.

Since LLMs are often used in more open-ended generation contexts, we will now explore the performance variance across prompt formats when considering sentence-length generation tasks (e.g. generate the next sentence of a story, given the four initial sentences of a story, generate a question whose answer is the sentence given). To analyze the automatic generations, we use two widely used metrics: ROUGE-L (Lin, 2004), and BERTScore (Zhang et al., 2019). The first is an n-gram-based metric, and the latter is a model-based metric, and both are $[0, 1]$ metrics where higher is better. Figure 19 shows that variance remains high for LLaMA-2-7B regardless of the metric and the number of n-shots considered, with LLaMA-2-7B 5-shot having 25% of tasks with a ROUGE-L spread of 0.098 or higher, and a BERTScore spread of 0.09 or higher.

We observe that the median spread is sometimes smaller than in the accuracy tasks. This may be because although ROUGE, BERTScore, and accuracy are all $[0, 1]$ metrics, typical metric values may be different, which may in turn affect the final spread (an absolute difference). We leave it to future work to quantify the differences in style or content that each format may be inducing.

Finally, it is worth noting that text generation metrics are known to be noisier, and thus not all metric decreases necessarily correspond to a true performance loss, as is the case for accuracy in single-valid-output tasks. We used 17 SuperNatural Instructions tasks: `task037`, `task038`,

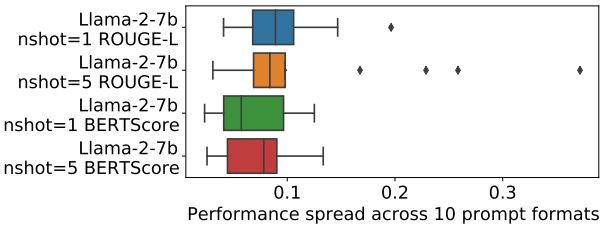

Figure 19: Spread across n-shots for LLaMA-2-7B, considering ROUGE-L and BERTScore metrics. 17 sentence-level open-generation tasks are considered, all extracted from SuperNatural Instructions. 10 prompt formats are considered for each task.

`task040`, `task067`, `task071`, `task072`, `task105`, `task216`, `task223`, `task240`, `task348`, `task389`, `task443`, `task845`, `task1326`, `task1401`, `task1613`. We selected the 17 open-ended text generation tasks among those with at least 1000 samples, with some formatting present in the original task (e.g. `'Passage:' <text>`). We only considered tasks whose instructions were under 1,000 characters and that contained inputs no longer than 5,000 characters.

We limit generations to 50 tokens. To parse model outputs more faithfully, and given that none of our expected generations include a newline, we only consider a model's generation up to the first newline (excluding leading spaces and newlines in a given generation). This consideration is important given that often models start to generate a new data sample from scratch, immediately after generating the requested answer.

Characterizing a model's accuracy distribution beyond spread. Spread gives a quantitative jump in information with respect to informing a single point in the performance distribution since it measures the distribution range (maximum minus minimum). However, distributions that may share the same range, may yield a widely different probability of obtaining each value in the distribution. Figure 20 plots the accuracy distribution of 30 tasks, sorted in decreasing order by standard deviation. Tasks with high standard deviation reflect a higher likelihood of obtaining dissimilar values when making a formatting selection; Figure 20 shows that the median standard distribution is $\sigma \approx 0.04$, which can be considered high in our context.

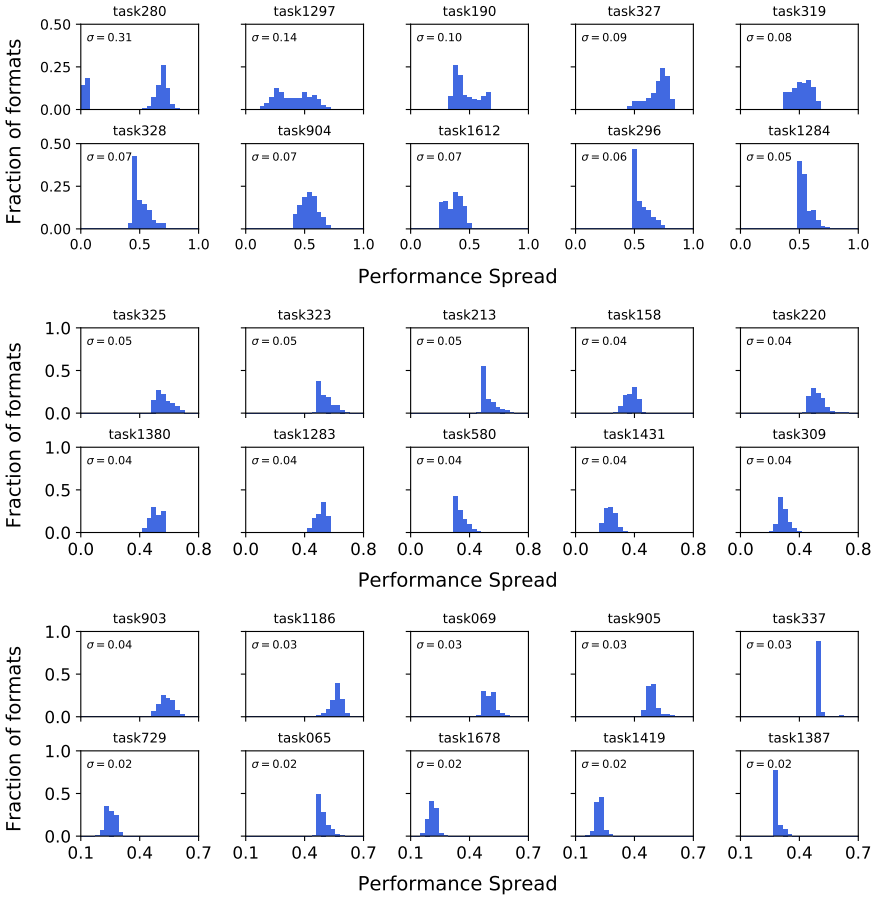

Figure 20: Accuracy distribution across 500 formats for 30 tasks evaluated on 250 samples each, sorted by standard deviation in decreasing order. LLaMA-2-7B 1-shot, option ranking metric.

On factors influencing spread besides prompt formatting. We believe many factors beyond formatting may be influencing performance variance, but were unable to find a feature that reliably predicts spread. We found that the average prompt length in a task has a negligible correlation with its performance spread: $r = 0.228$ ($p = 1.4 \times 10^{-7}$) for exact prefix matching metric, and $r = -0.022$ ($p = 0.615$) for option ranking metric, when jointly considering all models and n-shots. Similarly, the standard deviation of the prompt length had negligible correlation with spread: $r = 0.125$ ($p = 0.004$) for exact prefix matching, and $r = -0.099$ ($p = 0.024$) for option ranking metric. When considering each model individually, only LLaMA-2-7B with exact prefix matching showed a correlation $|r| > 0.5$, with the average prompt length having a correlation $r = 0.559$ $p = 6.86 \times 10^{-10}$. All other settings had $|r| < 0.36$.

B.3 PCA EXAMPLES

Section 4.4 systematically analyzes whether we can predict the prompt format that generated a given pre-softmax activation layer (i.e., prompt embeddings) by using solely its top-n principal components. Figure 21 shows the top two principal components for two different tasks where all 10 formats considered are easily identifiable solely with a prompt embedding's top two principal compoenents.

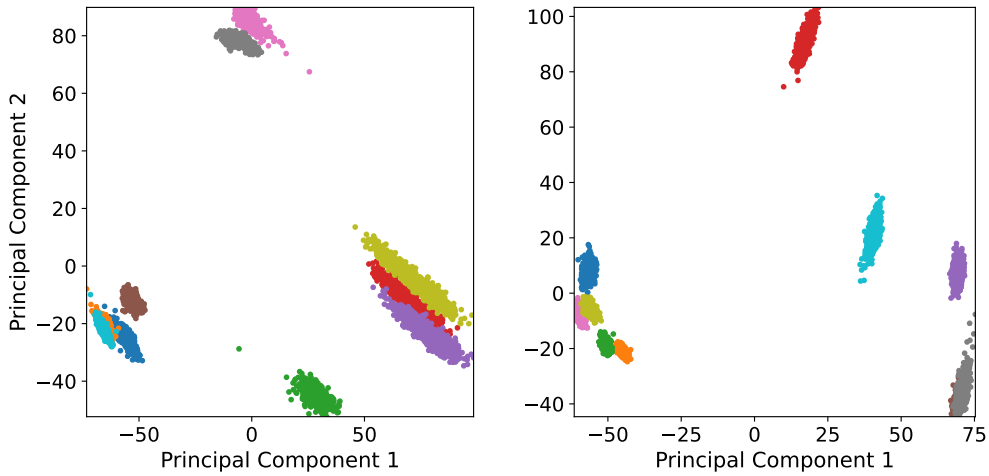

Figure 21: Plot of the top two principal components of the last decoder layer of the prompt, as a representation of the output probability distribution. Two different tasks shown, with each prompt format shown in a different color.

B.4 NOTABLE FEATURES

As discussed in Section 4.3, sometimes the choice of a constant may lead to significantly different accuracy ranges. Figures 22,23, and 24 show all strongly dissimilar choices of constants found on any given task, across 53 Super Natural-Instructions tasks, and on both accuracy metrics considered throughout the work. As can be appreciated, choices of constants do not consistently predict performance in isolation.